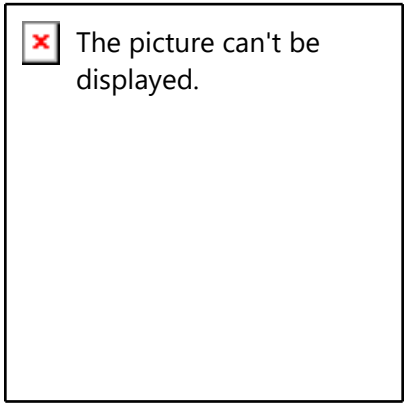
Toward a Standard Benchmark Format and Suite for Floating-Point Analysis



Nasrine Damouche, Matthieu Martel, **Pavel Panchekha**, Chen Qiu, Alexander Sanchez-Stern, Zachary Tatlock.

Incredible progress…

Automatic Verification
Fluctuat [SAS'13]
Rosa [POPL'14]
FPTaylor [FM'15]

Optimization
STOKE [PLDI'14]

Improvement
Salsa [FMICS'15]
Herbie [PLDI'15]

Mechanized Proofs
Wave equation [ITP'10]
Rounding error [NSV'16]

Rapid improvement in hard problems!

Incredible progress…

We want our community
to keep progressing!

Next
???

Rapid improvement in hard problems!

Incredible progress

We want our community to keep progressing!

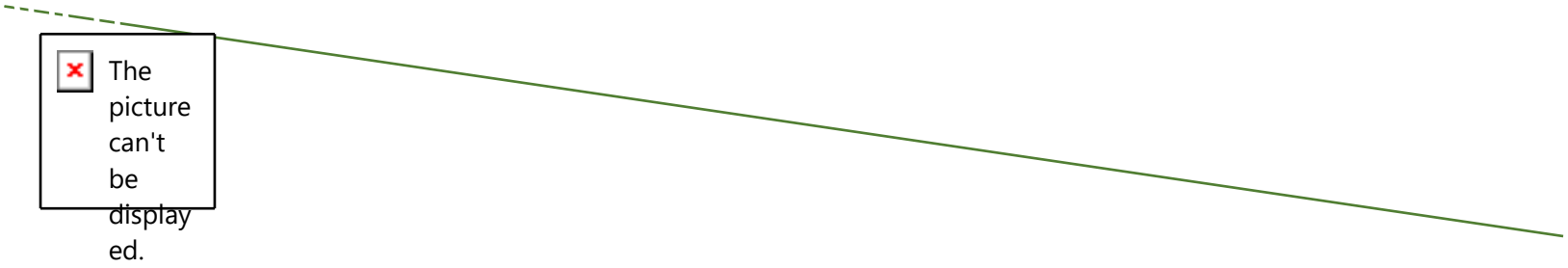
Auto
Fluc
Rosa
FPTa

Next
???

Salsa
Herbie

As community grows, growing pains appear

Rapid improvement in hard problems!



FPBench is **community infrastructure** for **cooperation** and **comparison** in the FP community.

http://fpbench.org

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FPBench

β

Common format

Benchmark suite

Named measures

The picture can't be displayed.

FPBench

β

Common format

Benchmark suite

Named measures

$$\sqrt{x+1} - \sqrt{x}$$

Arguments

```
(FPCore (x)
```

```
 (- (sqrt (+ x 1)) (sqrt x)))
```

S-expression syntax

```
(FPCore (x0)
  :name “Sine Newton”
  :cite (darulova-kuncak-2014)
  :pre (< (abs x0) 1)
  (while (< i 10)
    ([i 0 (+ i 1)]
     [x x0 (+ (+ (- x (/ (pow x 3) 6.0))
                 (/ (pow x 5) 120.0))
                 (/ (pow x 7) 5040.0))
              (+ (+ (- 1.0 (/ (* x x) 2.0))
                 (/ (pow x 4) 24.0))
                 (/ (pow x 6) 720.0))))])
    x))
```

Loops

Common functions

FPCore common format

Simple to use

S-expression syntax
Purely functional
No control flow analysis

Expressive

All C, Fortran functions
Loops, conditionals
Tools support parts

Extensible

Metadata properties
Tool-specific metadata
Input or output format



Generate from higher-level, imperative FPImp lang.

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FPBench

β

Common format

Benchmark suite

Named measures

Simple to implement

Covers all existing uses

Simple to extend, specialize

The picture can't be displayed.

FPBench

β

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Simple to extend, specialize

FPBench benchmark suite

Existing programs	Rich features	Diverse domains
FPTaylor	Arith	Textbook
Herbie	Expt	Math Alg
Rosa	Trig	Emb Sys
Salsa	Loop	Sci Comp
	Branch	

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FPBench

β

Common format

Simple to implement

Covers all existing uses

Simple to extend, specialize

Benchmark suite

From existing projects

Cover many domains

Grows over time

Named measures

The picture can't be displayed.

FPBench

β

Common format

Simple to implement

Covers all existing uses

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Benchmark suite

From existing projects

Cover many domains

Grows over time

Named measures

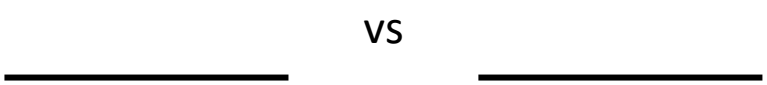
FPBench measures

Formal definitions of accuracy measures

Described along 5 axes

Standard measures so tools agree

FPBench axes of measurement

Scaling vs. non-scaling	Absolute, relative, ULPs, bits, …
Forward vs. backward	Fixed input error vs fixed output error
Maximum vs. average	
Sound vs. statistical	Formal guarantees vs mathematical accuracy
Improvement	

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FPBench

β

Common format

Simple to implement

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Benchmark suite

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Cover many domains

Grows over time

Named measures

Terms for measuring error

Standard across tools

Flexible but rigorous